NATQ2-DIAG-002 — pre-CE trace reconstruction

Diagnostic only · 0 systems evaluated · CE inference False · validation_runs_consumed 1, 2 unused · reserve 0 bytes · head `88d98b4` · 2026-09-05 19:31Z

This corrects DIAG-001's emphasis. That report named the cross-encoder and stopped. The reconstructed pre-CE ordering shows the retrieval stage was weak on the same spans: **only 3 of the 12 ranked-out gold spans were inside the pre-CE retrieval top 10**, and **9 sat beyond rank 20**, as deep as 68. So there is no strong retrieval signal to protect. A guard keyed on the existing ordering rescues at most those 3 cases — 23/40 → 26/40 = 0.65, *exactly the FAIL floor* and nowhere near the 0.80 PASS floor. That rules out the retrieval-protected reranker and is why the recommendation below is **B**, not A.

EXACT	**3/12**	**9/12**	**4,852**	**2**
REPLAY IDENTITY	PROTECTED BY RETRIEVAL	PRE-CE RANK > 20	CANDIDATES TRACED	RUNS STILL UNUSED

1 — Replay identity

19 hash-pinned implementations from `origin/grok/v2-natq-20260903`. No cross-encoder was invoked; every CE score is the persisted EVAL-NATQ2-H-002 value, and the join was required to be a bijection on every case — it was (True).

Reproduction check	Mismatches
Final top-10 rows (400)	0
Case hit vector (40)	0
Per-case span ranks	0
Aggregate metrics	identical

The final ranking reproduced exactly even though non-top-10 CE scores were stored at 6 decimals, which is what establishes that the rounding never flipped an ordering at the depth-10 boundary.

2 — Pre-CE rank of the gold spans

pre-CE retrieval rank	12 ranked out	all 44 in pool	32 in final top 10
<=3	0	17	17
<=5	2	5	3
<=10	1	9	8
<=20	0	2	2
>20	9	11	2

RETRIEVAL_TOP10_TO_FINAL_OUT = 3/57 spans, 3/40 cases (B28, D27, E06) — all three finished at final rank 11, 11 and 14, just outside. This is a mechanism measurement, not a system result.

case	span	origin	A rank	local	pre-CE	CE score	CE rank	final
A07	1	a_pool	32	—	56	-2.116	63	68
A30	0	a_pool	40	—	56	-2.268	18	31
B09	0	a_pool	11	—	26	-6.37	20	20
B09	1	a_pool	11	—	26	-6.37	20	20
B22	0	a_pool	47	—	68	-7.578	50	65
B28	0	a_pool	4	1	4	-4.518	17	11
C03	0	a_pool	38	—	59	-2.389	31	44
C09	0	a_pool	11	—	25	0.096	19	20
D03	0	a_pool	80	7	22	-4.226	17	19
D08	0	a_pool	14	—	46	-3.351	7	15
D27	0	a_pool	4	1	4	-3.378	32	11
E06	0	a_pool	7	9	10	-2.796	30	14

3 — Retrieval-only versus SYSTEM-H

> `RETRIEVAL_ONLY_TOP10_HIT` is an internal mechanism diagnostic. **It is not a qualified system** and must never be reported as one.
>
> retrieval hit / H hit **21** · retrieval hit / H miss **3** · retrieval miss / H hit **2** · retrieval miss / H miss **14**
>
> Diagnostic rate 24/40 against SYSTEM-H’s closed 23/40: the CE/blend stage trades 3 cases away (B28, D27, E06) and buys 2 back (C11, D29). Near-parity does not make an ordering a system.

BM25 regressions. B09 **RETRIEVAL_AND_CE_BOTH_WEAK**. B28 **RETRIEVAL_ALREADY_CORRECT_CE_DESTROYED**. D27 **RETRIEVAL_ALREADY_CORRECT_CE_DESTROYED**. B09 sat at SYSTEM-A rank 11, pre-CE rank 26 and CE −6.37 — both stages weak. B28 and D27 were both at pre-CE rank 4 and pushed to final rank 11.

4 — Channel contribution

population	BOTH_A_AND_LOCAL	SYSTEM_A_ONLY	LOCAL_BM25_ONLY	PROJECTION_ONLY
in pool (44)	35	9	0	0
final top 10 (32)	31	1	0	0
ranked out (12)	4	8	0	0

Corroboration across both retrieval channels tracks survival closely: 31 of the 32 surviving gold spans were found by SYSTEM-A *and* local BM25, while 8 of the 12 ranked-out spans were SYSTEM-A only. Local BM25 contributes a mean 10.0 additive candidates out of 99.2 available.

5 — Projection is inert, not harmful

800 projection candidates across the 40 queries covered **0** gold spans. 12 entered the final top 10 across 8 queries, a mean of 0.3 slots per query. For **every one** of those 12 slots, the next non-projection candidate in frozen final score order covers no gold span —

0 of 12. Projection displaced nothing gold. **No removal was simulated and no claim is made that removing it improves any metric.**

6 — Localization, not document discovery

Of the 13 gold spans with no covering candidate, **12** had the gold document present in SYSTEM-A and only **1** was absent from the pool entirely. But only **6** had that document among the local-BM25 parents, so the localization stage was not pointed at the right document for the other seven. Projection windows covered the gold chunk in 0 of 13.

case	span	doc in SYSTEM-A	doc in local parents	same-doc candidates	nearest final rank
A12	0	True	True	6	13
A12	1	True	True	6	13
A12	2	True	True	6	13
A30	1	True	False	1	31
B18	0	True	False	4	40
B19	0	True	True	7	2
B19	1	True	True	7	2
C01	0	True	False	2	57
C01	1	True	False	2	57
C03	1	False	False	0	None
C05	0	True	False	1	101
E01	0	True	True	38	1
E27	0	True	False	3	45

E01 is the clearest shape: 38 candidates from the gold document sit in the pool, the nearest at final rank 1, and none of them covers the gold span. B19 is the same at rank 2. That is a chunk-boundary problem, not a retrieval problem.

7 — Recommendation: B. NEW RERANKER MODEL

> The evidence is in the pool (44/57) but the frozen retrieval ordering offers no useful protection: only 3 of the 12 ranked-out spans were in the pre-CE top 10 and 9 were beyond rank 20.

- **Why not A.** A guard on the existing retrieval ordering rescues at most the 3 RETRIEVAL_TOP10_TO_FINAL_OUT cases, taking case_hit@10 to 26/40 = 0.65 — exactly the FAIL floor.
- **Why not C.** Ranking still has room: 23/40 against a 33/40 ceiling is 10 cases and the floor needs 9.
- **Why not D.** One mechanism dominates — the reranker misorders evidence it already holds.

> **The qualification caveat stands.** The any-span ceiling is 33/40 = 0.825 against a 32/40 = 0.80 floor: one case of oracle margin. A ranking fix alone is not sufficient evidence for reserve. The ceiling work has a target: parent selection for the localization stage, which reaches the gold document for only 6 of the 13 missing spans.

Candidate ceilings reconfirm DIAG-001 exactly — any-span 33/40, every-span 31/40, span 44/57: True.

8 — Artifacts

File	SHA256
NATQ2-DIAG-002-REPLAY-IDENTITY.json	a55843cd9b67801c6ba58141328bcf520e28ea9560da76d43d25ef1359a40d46
NATQ2-DIAG-002-CANDIDATES.jsonl	1961a5d6b319ea69c39979d911722d6cf483b21eaa5ca637f40641d31a245c1c
NATQ2-DIAG-002-GOLD-SPANS.json	1f2a16fc99ba21f2edc596ca1addcb4bd69346ce02e662f1eaacf5e854bee834
NATQ2-DIAG-002-AGGREGATE.json	2c47cb67b8232219fd568e971ab438334fe70fc472c051289dcbeccd875ac4bb
NATQ2-DIAG-002-REPORT.md	e0448ccb73a293abf101dfdda987b48ae2f6477d81b60a5954043ed5c623b2ec

SYSTEM-H config hash `7599eb3c3bb4798230a722e6a7c96b046dc36cc0332b584ec55339896b2d717a`, unchanged. SYSTEM-H validation remains CLOSED as FAIL and was not rerun; no validation run was consumed by this task. No parameter search: no guard K tested, no blend weight changed, no projection removed, no retriever run, no candidate pool altered, no successor implemented. Reserve verified before and after: 0 bytes, reserve_frozen True, reserve_count 60. Prior status at head `88d98b4`.